

SAN FRANCISCO BUOY 46026, CA, USA

WMO: 992640

Lat: 37.754N

Lon: 122.839W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (NAP)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.6	8.3	N/A	N/A	N/A	N/A	N/A	N/A	16.4	9.8	15.4	9.7	6.2	320	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	1.8	17.7	N/A	16.9	N/A	16.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	13.2	11.1	DB	6.6	19.7	1.3	1.2	5.6	20.6	4.8	21.3	4.1	22.0	3.1	22.9	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	11.3	11.3	11.0	10.9	11.1	11.7	12.8	13.7	13.9	13.5	12.6	11.5
	DBStd	1.75	1.44	1.47	1.40	1.11	1.23	1.07	1.37	1.52	1.44	1.48	1.38	1.53
	HDD10.0	23	4	4	5	4	2	0	0	0	0	0	0	3
	HDD18.3	2270	218	197	226	224	224	199	172	145	134	149	171	211
	CDD10.0	796	44	41	37	30	37	51	87	113	116	109	79	51
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	5.7	6.2	6.3	6.5	6.6	6.3	4.9	4.8	4.6	5.0	5.4	6.0
Precipitation	PrecAvg	796	163	163	87	48	31	6	0	2	2	37	81	177
	PrecMax	1240	424	479	278	171	104	26	1	24	9	134	254	467
	PrecMin	400	17	7	1	3	0	0	0	0	0	5	16	
	PrecStd	222	117	104	83	40	30	8	0	5	2	36	70	128
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	15.8	16.2	14.8	16.5	15.3	17.0	18.4	18.2	18.5	17.7	16.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	14.3	14.9	14.2	13.4	14.6	14.6	16.3	17.5	17.4	17.0	15.9	15.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	13.8	14.0	13.5	12.8	13.5	14.0	15.6	16.7	16.6	16.4	15.2	14.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	13.3	13.2	12.9	12.4	12.8	13.3	15.0	15.9	15.8	15.7	14.7	13.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.1	2.0	1.8	1.5	1.5	1.5	1.6	1.7	1.8	2.0	1.9	2.1
		MCDBR	2.5	2.9	2.7	2.1	2.5	1.9	1.8	1.9	2.2	3.1	2.6	2.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.322	0.331	0.340	0.357	0.364	0.356	0.368	0.340	0.343	0.337	0.321	0.322	
	taud	2.575	2.548	2.523	2.439	2.447	2.474	2.448	2.533	2.500	2.560	2.584	2.579	
	Ebn at Noon	868	904	926	922	916	920	906	928	908	884	862	841	
	Edh at Noon	75	86	97	111	113	110	112	100	98	84	73	70	
All-Sky Solar Radiation	RadAvg	2.37	3.30	4.64	6.05	6.82	7.36	6.60	5.85	5.18	3.91	2.78	2.06	
	RadStd	0.32	0.40	0.40	0.45	0.51	0.38	0.34	0.37	0.30	0.27	0.19	0.29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (41 neighbors)	N/A	N/A	-1.21	N/A	N/A	N/A	N/A	-108	+135	N/A

Nomenclature: See separate page